

PhD Lecture in Behavioral Asset Pricing: Preferences & Beliefs (II)

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Psychology of Beliefs

Psychology of Beliefs: Representativeness

- Consider questions such as:
 - What is the probability that object A comes from class B?
 - What is the probability that event A was generated from process B?
- Kahneman and Tversky (1974) argue that people often answer by using the “representativeness heuristic”
 - Evaluate the probability by the extent to which A is representative of B
 - i.e. degree to which A reflects the essential characteristics of B
- This is helpful in general but can lead to serious biases

- **Base-rate neglect**
- Consider the following description of Steve. Do you think Steve is more likely to be a librarian or lawyer?
 - Steve is very shy and withdrawn, invariably helpful, but with little interest in people or in the world of reality.
 - A meek and tidy soul, he has a need for order and structure, and a passion for detail.
- People judge Steve to be more likely a librarian than a lawyer
- But there are much more lawyers than librarians in the U.S.
- The question actually ask you about the LHS while people answer it with the first formula of RHS; the root cause of base rate neglect.

$$\frac{p(\text{library} \mid \text{data})}{p(\text{lawyer} \mid \text{data})} = \frac{p(\text{data} \mid \text{library}) p(\text{library})}{p(\text{data} \mid \text{lawyer}) p(\text{lawyer})}.$$

- **Sample size neglect**
- When we do not know the model generating the data, we may infer it too quickly based on too little data
- This can lead to over-extrapolation of recent trends
- E.g., the hot hand fallacy
 - Seeing a basketball player make a few shots in a row leads to the belief that he is “hot”
 - But empirical evidence suggests that the hot hand is largely a myth (Gilovich, Vallone, and Tversky, 1985)

Psychology of Beliefs: Overconfidence

- There are at least two distinct types of overconfidence:
- **Overprecision**
 - People are too confident in the accuracy of their forecasts
 - 98% confidence intervals include the true quantity around 60% of the time (Alpert and Raiffa, 1982)
- **Overplacement**
 - People think they are better than others
 - 90% of drivers think they are above-average drivers (Svenson, 1981)

Overprecision: Survey Evidence

- Ben-David, Graham, and Harvey (2013)
- CFOs' expected stock market return distribution is too narrow
- Realized market returns are within the executives' 80% confidence intervals only 36% of the time

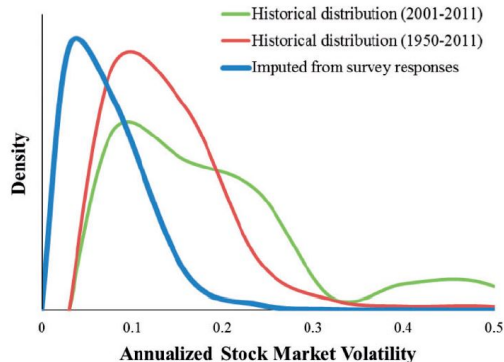


FIGURE IIA

One-Year Imputed Individual S&P 500 Volatility Versus Historical Distribution

The kernel density of the one-year S&P 500 return volatility for the period of 1950–2000 (or 2001–2011), and the kernel density of the imputed one-year S&P 500 individual volatility derived from CFOs' predictions.

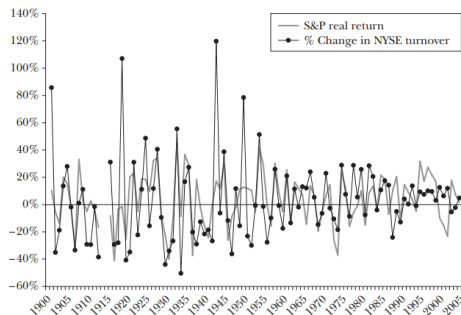
Overconfidence: Excessive Trading

- **Excessive trading**

- Under efficient markets, there is no trading other than for liquidity needs and portfolio rebalancing
- Speculative trading reflects differences of opinion and some amount of overconfidence
- Trading leads to lower returns
 - The more you trade, the lower your return (Odean, 1999)

Figure 3

Market-Wide Stock Returns and Changes in Turnover, 1901–2005



Hong and Stein (2007)

Overconfidence: Excessive Trading

- **Boys will be boys**
- Barber and Odean (2001) show that men trade more frequently than women, but earn lower returns
- Consistent with psychological studies that men are more overconfident than women (Lundeberg et al., 1994)

Dependent variable	Mean monthly turnover (%)	Own-benchmark abnormal net return	Portfolio volatility
Intercept	6.269*** (4.24)	-0.321*** (-11.47)	11.466*** (58.85)
Single	0.483*** (4.24)	0.002 (0.14)	0.320*** (3.40)
Woman	-1.461*** (-12.76)	0.058*** (4.27)	-0.689*** (-7.27)
Single × woman	-0.733*** (-3.38)	0.027 (1.08)	-0.439*** (-2.45)



Overconfidence: Momentum

- Daniel, Hirshleifer, and Subrahmanyam (1998)
- Momentum is driven by investor overconfidence and self-attribution bias
 - Investors are overconfident in the private information they gather
 - Investors also suffer from self-attribution bias: they credit themselves for successes but blame external factors for failures
- Positive returns reinforce overconfidence, leading to more aggressive trading and price momentum
- Eventually, prices correct when information is revealed

Overconfidence: Momentum

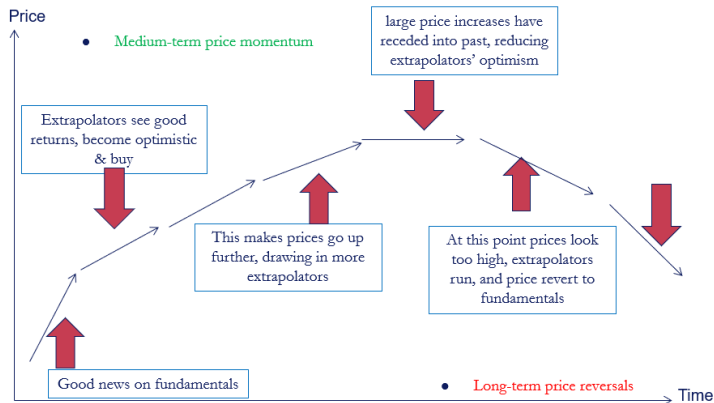
- Chui, Titman, and Wei (JF 2010) test the idea that momentum is driven by investor overconfidence and self-attribution bias
- They use an individualism index developed by Hofstede (2001) to proxy for investor overconfidence
- They find that momentum is stronger in countries with higher individualism scores

Portfolio Formed Method	Index on Individualism	Winner (W)	Loser (L)	W Minus L
Panel A: Country-Average Portfolios				
Country-average	Low	1.628 (4.30)	1.241 (2.96)	0.387 (2.80)
	2	1.693 (5.22)	1.004 (2.90)	0.689 (5.91)
	High	1.748 (6.14)	0.707 (2.27)	1.041 (7.93)
	High minus low	0.120 (0.39)	-0.534 (-1.53)	0.654 (4.30)

Psychology of Beliefs: Over-extrapolation

- The estimate of the future value of a quantity is a positive function of the recent past values of that quantity
- One of the most important and widely-applied ideas in behavioral finance
- The source of over-extrapolation: the representativeness heuristic
- Application to finance:
 - Return extrapolation: predict future returns based on past returns
 - Fundamentals extrapolation: predict future earnings/cash flows based on past growth

Over-extrapolation: Conceptual Framework

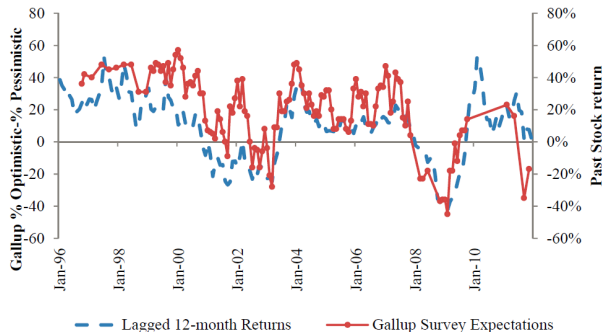


Over-extrapolation: Survey Evidence

- **Excess volatility and predictability in aggregate stock market**
- Extrapolators buy assets whose prices have risen because they expect them to keep rising
- This leads to overvaluation and return predictability

The Role of Past Stock Market Returns in Explaining Survey Expectations

The dashed line denotes the 12-month rolling nominal return on the CRSP VW stock index. The solid line marked with circles denotes expectations from the Gallup survey (% optimistic - % pessimistic).



Greenwood and Shleifer (2014)

Over-extrapolation: Survey Evidence

- Investors act on their over-extrapolative beliefs with flows into (out of) stock markets

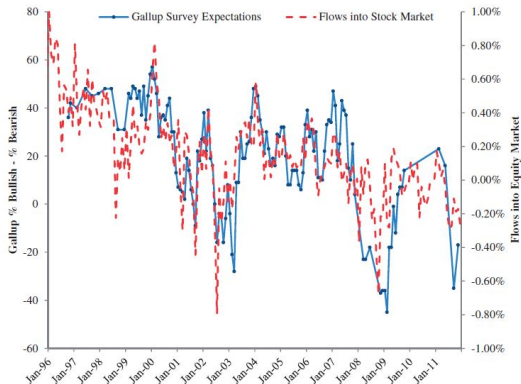


Figure 4

Comparing the Gallup survey with flows into equity mutual funds

The solid line denotes the percentage of investors who are bullish in the Gallup survey (left axis). The dashed line (right axis) is flows into mutual funds as a percentage of equity market capitalization, as reported by the Investment Company Institute.

Greenwood and Shleifer (2014)

Over-extrapolation of Past Returns: Applications

- Using survey data on expectations of stock returns, Casella and Gulen (RFS 2018) estimate the degree of extrapolative thinking in investors' beliefs (DOX) about market returns
- DOX $(1 - \lambda)$ measures the weight investors place on past returns

$$Exp_t = a + b \sum_{i=0}^{\infty} w_i R_{t-(i+1)\Delta t, t-i\Delta t},$$

$$w_i = \frac{\lambda^i}{\sum_{k=0}^{\infty} \lambda^k}, \quad 0 \leq \lambda < 1,$$

- A lower λ implies that investors place higher weight on more recent observations
- Earlier observations contribute less to an extrapolator's expectations

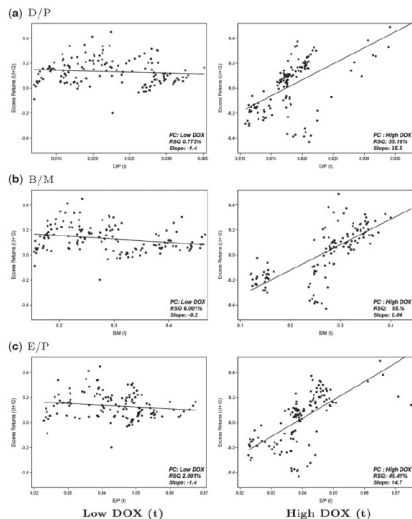


Figure 2
Time-varying DOX

The figure plots the recursively estimated DOX time series for the principal component (PC) of the II and AA surveys, in the sample period 1992:06–2013:12. (Details on estimation are in Section 2.)

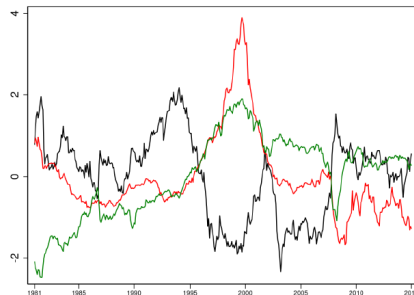
Over-extrapolation of Past Returns: Applications

- They find the predictability of stock returns by a price-scaled variable (e.g., D/P, B/M or E/P) is contingent on the DOX
- When DOX is high, investors more quickly “forget” the positive price changes that drove up valuations, and their beliefs mean revert faster
- As a result, high valuations are more strongly associated with low future returns when DOX is high



Over-extrapolation of Past Fundamentals: Application

- Bordalo et al. (2023): the overreaction of long-term earnings growth (LTG) expectations helps explain time-series variation in the P/D ratio and predicts aggregate market returns
- Good news cause investors to become overly optimistic about future earnings growth



- This figure graphs LTG (red) and the price dividend ratio (green)
- along with realized stock returns over the coming five years (black)

My Research: Naïve Earnings Growth Extrapolation

- **Li, Cui, and Zhang (R&R, Review of Accounting Studies)**

- China reports quarterly earnings on a year-to-date basis, which makes cumulative YTD growth highly salient, even though this YTD measure contains stale earnings information
- Stocks with the strongest past YTD earnings growth run up in the five days before announcements and reverse afterward
- investors extrapolate the salient YTD growth into the upcoming earnings announcement
- The pattern is stronger for small, retail-heavy, low-coverage stocks, and retail buying demand rises before announcements

- **Connection:** the cross-sectional counterpart of Bordalo et al. (2023) on the previous slide: fundamentals extrapolation that distorts individual stock prices around earnings news, not just the aggregate market

Psychology of Beliefs: Conservatism

- Investors with extrapolative or overconfident beliefs adjust their views too much in response to new information and cause excessive movements in prices
- But some phenomena — momentum and PEAD — may instead require that investors adjust their views *too little*
- **Conservatism:** Individuals are too slow to update their beliefs in response to new information (Edwards, 1968)

Psychology of Beliefs: Belief Perseverance and Confirmation Bias

- **Belief perseverance:**

- Once we have formed an opinion, we are often too slow to change it on receipt of new evidence
- We do not look for evidence that would falsify our beliefs
- We ignore evidence that goes against our beliefs

- **Confirmation bias:**

- We actively seek out evidence that supports our beliefs
- We interpret ambiguous evidence as supportive of our beliefs

Psychology of Beliefs: Anchoring

- Kahneman and Tversky (1974): in forming numerical estimates of uncertain quantities, adjustments in assessments away from some initial value (called the “anchor”) are often insufficient
 - E.g. estimates of the percentage of countries in the United Nations that are African countries were affected by a number between 0 and 100 that was determined by spinning a wheel of fortune in the subjects’ presence
 - Those who were asked to compare their estimate to 10, subsequently estimated 25 percent, while those who compared to 60, estimated 45 percent
- Applications in Finance:
 - Anchoring on 52-week high in stock market George and Hwang (2004): nearness to the 52-week high predicts future return, and argue this is because traders are slow to react to good news when price is close to the 52-week high
 - This also explains the momentum effect

The Effect of Feelings

- A recent strand of research investigates how feelings affect beliefs and preferences
 - Affective feelings (mood, emotion)
 - Cognitive feelings
- **Affective feelings**
 - A happier mood caused by an exogenous stimulus leads to more optimistic judgments and choices
 - A sadder mood leads to more pessimistic judgments
 - E.g. sunshine, sports outcomes, holidays

Affective Feelings: Applications

- **Sports**

- Edmans, Garcia, and Noli (2006): When the national soccer team loses a World Cup match, the national stock market falls the next day
- Disappointing sports news worsens the national mood, leading people to form more negative judgments about something unrelated, namely the economy's prospect
- The effect is stronger in countries where soccer is more important

- **Sunshine/Daylight**

- Hirshleifer and Shumway (2003): stock market has higher returns on sunnier days
- Kamstra, Kramer, and Levi (2003): fewer hours of daylight lead to a type of depression, Seasonal Affective Disorder (SAD), which, in turn, leads to risk aversion and lower return

- **Terrorist attacks**
- Wang and Yang (2020): Terrorist attacks lead to increased aggregate risk aversion, as proxied by flows into (out of) government bond funds (equity funds)
- These findings point to a more general mechanism that may amplify market fluctuations
 - If the stock market goes up, this improves the mood of many investors, leading them to form more positive judgments about the future prospects of the economy
 - This causes them to push the stock market up still higher

My Research: Air Pollution and Carbon Divestment

- **Li, Huynh, and Xia (RAST 2025)**

- Mutual fund managers exposed to local air pollution underweight stocks of high-emission firms
 - The effect strengthens for environmentally conscious managers and for those surprised by pollution: managers revise their carbon-risk beliefs from an environmental stimulus
 - The divestment is costly: sold carbon-intensive stocks subsequently outperform the stocks bought in their place
- **Connection:** like soccer losses and terrorist attacks, pollution is an exogenous stimulus that shifts beliefs about environmental risks, and here professionals *overreact* by selling carbon-intensive stocks more than justified by fundamentals

Cognitive Feelings

- The feeling of ease when accessing or processing information leads to more positive judgments than is justified by content alone (Zajonc, 1968; Bornstein, 1989)
- E.g. people like familiar things more than justified by content alone
- In particular, mere exposure to something makes us like it more than justified
- **Applications:**
 - Home bias in portfolio choice
 - Local bias in stock ownership
 - Excessive allocation to own company stock
- We are exposed, on a daily basis, to our home country, our local region, and the company we work for

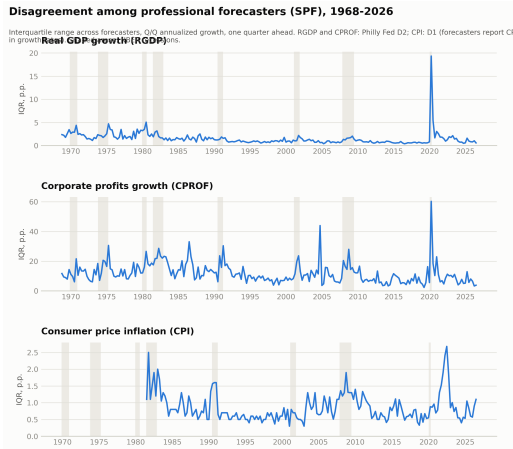
Heterogeneous Beliefs

Heterogeneous Beliefs: Overview

- Previously we looked at the psychological foundations of systematically biased beliefs
 - Assuming representative investor whose belief is biased
- In this part, we look at models with heterogeneous beliefs
 - Average belief is unbiased
 - But cross-sectional dispersion in beliefs can still affect prices and trading

Heterogeneous Beliefs, ctd.

- People regularly hold different beliefs about almost everything in life
 - from political views and sport events to economic outcomes and asset returns

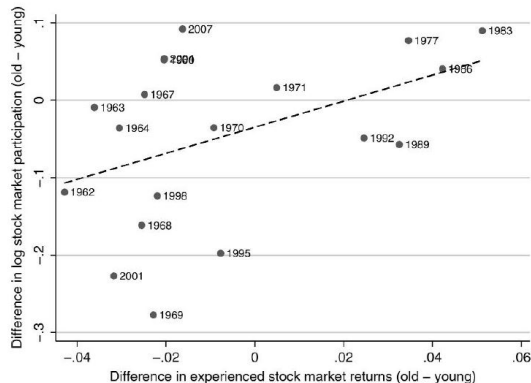


Sources of Heterogeneous Beliefs

- What may cause heterogeneous beliefs?
- Asymmetric information alone cannot cause rational agents with common prior to agree to disagree
 - e.g., No-trade Theorem of Milgrom and Stokey (1981)
- Heterogeneous beliefs must come from:
 - Differences in prior beliefs: It is reasonable for agents to hold different priors about new things, e.g., Morris (1995)
 - Differences in how agents process information: investors put different weights on the same piece of information
 - The root cause can come from overconfidence, culture or experience

Sources of Heterogeneous Beliefs — Experience Effects

- One important source of heterogeneous beliefs is personal experience
- When forming beliefs, a rational individual should take account of all available information
- However, mounting evidence suggests that people overweight their personal experience
- Malmendier and Nagel (2011): whether a household participates in the stock market, and how much it invests in it, depends on the stock market return it has experienced in his/her lifetime



Sources of Heterogeneous Beliefs — Experience Effects

- **Inflation expectation** (Malmendier and Nagel, 2016)
- Individuals overweight inflation experienced during their lifetimes when forming expectations about future inflation
- Younger individuals update their beliefs more strongly than old individuals
- Kuchler and Zafar (2019) show that people's beliefs about aggregate economic outcomes strongly depends on their locally experienced economic conditions (housing prices and unemployment)

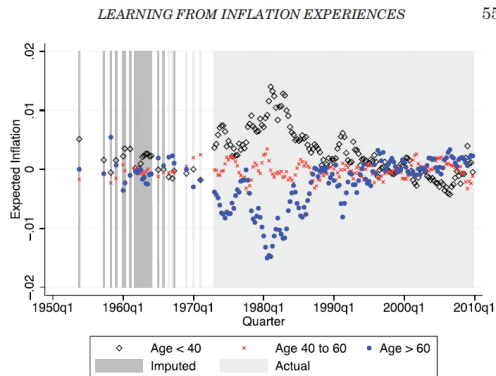


FIGURE I

Inflation Expectations by Age Group Relative to Cross-Sectional Mean

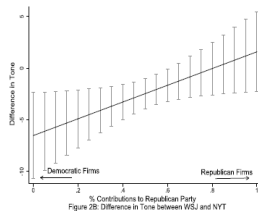
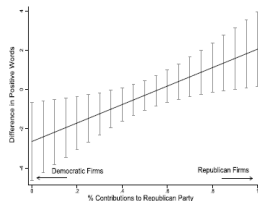
Malmendier and Nagel (2016)

Sources of Heterogeneous Beliefs — Political Polarization

- Goldman, Gupta, and Israelsen (JFE 2025)
- Setting: Comparing coverage of the same corporate financial news by the conservative Wall Street Journal and the liberal New York Times

- **Findings:**

- Newspapers use more positive language and are more likely to report good rather than bad news about politically aligned firms
- Politics-induced disagreement in corporate financial news leads to an increase in abnormal trading volume for the most politically extreme firms

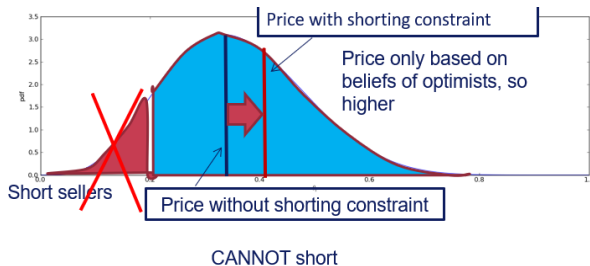


Heterogeneous Beliefs (no SSC)

- Taken alone, overconfidence-based heterogeneous beliefs do not have interesting pricing effects
- The pricing impacts of optimistic and pessimistic investors cancel out
- But they can help explain the high trading volume
 - Investors trade because they disagree about asset values
 - Higher dispersion of beliefs $\Rightarrow$ higher trading volume
- Grinblatt and Keloharju (2009):
 - Use data from Finland to show that more overconfident people trade more
 - Overconfidence is self-reported confidence minus how confident you should be based on performance on aptitude tests

Heterogeneous Beliefs with SSC

- If we combine heterogeneous beliefs with short-sale constraints, we get interesting pricing effects
- **Static argument**
- Miller (1977): Heterogeneous Beliefs + Short-Sale Constraint leads to overpricing

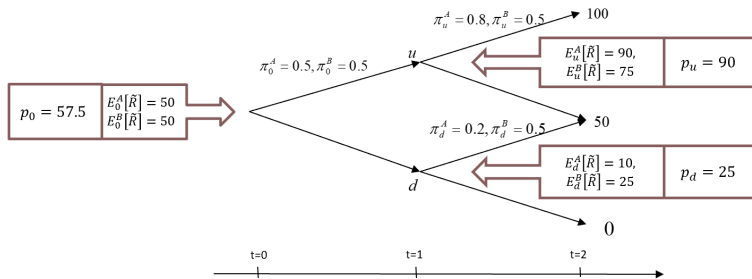


Heterogeneous Beliefs with SSC, ctd.

- **Dynamic argument**
- Harrison and Kreps (1978)
- If there are heterogeneous beliefs, you are willing to pay more than fundamental value today
 - When information is released, there is a good chance that you will be able to sell to someone more optimistic
 - This creates a “speculative” component to the price
- Scheinkman and Xiong (2003) build on this idea and develop “resale option theory” of speculative bubbles
- When short-sales are constrained, an asset owner values the option to resell asset to other future, more optimistic buyers

Heterogeneous Beliefs with SSC, ctd.

- **An example:** $t = 0, 1, 2$
- Risk-neutral investors A and B trade a risky asset
- Interest rate is zero and there is no short selling



Heterogeneous Beliefs with SSC, ctd.

- Both agents' buy-and-hold valuation of the asset is 50
- What is the traded price?
 - In state u , A sets the price at 90
 - In state d , B sets the price at 25
 - Then, at time 0, both value the asset at 57.5!
- This is because both anticipate to resell to the other agent at a higher price in the future.
 - B expects to resell to A in state u at 90, which is higher than his valuation 75.
 - A expects to resell to B in state d at 25, which is higher than his valuation 10.
- This resale option contributes to both A and B's initial asset valuation: A speculative bubble component.

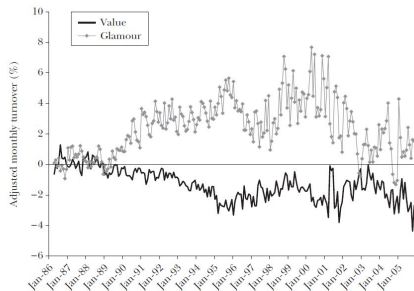
Heterogeneous Beliefs with SSC, ctd.

- Characteristics of this bubble component:
- The valuation of the resale option is determined by fluctuations of agents' beliefs and asset float
- Scheinkman and Xiong (2003): the more likely their beliefs flip, the more they trade and the more valuable the resale option.
- This is why bubbles tend to coincide with trading frenzies.
- Another key factor is asset float: More shares available for trading $\Rightarrow$ smaller bubble
 - This is because when investors have downward sloping demand curve, an owner can sell his share only if buyer's belief passes his by a sufficient margin
 - A larger float makes this less likely to happen

Heterogeneous Beliefs with SSC, ctd.

- Models of heterogeneous beliefs with SSC are popular because they not only explain overpricing, but also another important empirical fact:
- **The coincidence of high valuations and heavy trading**
- Hong and Stein (2007): Value stocks vs. growth stocks divergence in both the valuation and turnover during tech bubble period

Figure 2
Turnover in Value and Glamour Stocks, 1986–2005



Heterogeneous Beliefs and Bubbles

- Xiong and Yu (AER 2011) conduct a more rigorous test of the heterogeneous beliefs-based bubble theory — The Chinese Warrant Bubble
- In 2005–2008, over a dozen put warrants traded in China went so deep out-of-the-money that their intrinsic value was essentially zero
- Yet they traded at high prices with enormous turnover

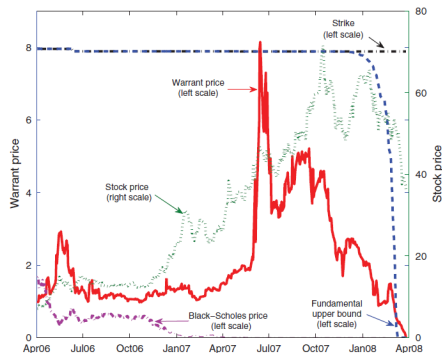


FIGURE 1. PRICES OF WULIANG PUT WARRANT

Notes: This figure shows the daily closing prices of WuLiang stock and its put warrant, along with WuLiang warrant strike price, upper bound of its fundamental value assuming WuLiang stock price drops 10 percent every day before expiration (maximum allowed per day in China's stock market), and its Black-Scholes price using WuLiang stock's previous one-year rolling daily return volatility.

Heterogeneous Beliefs and Bubbles

- They find evidence most consistent with the resale option theory of bubbles
- The size of the warrant bubble increases with turnover and volatility, and decreases with asset float

TABLE 5—DETERMINANTS OF SIZE OF THE WARRANTS BUBBLE

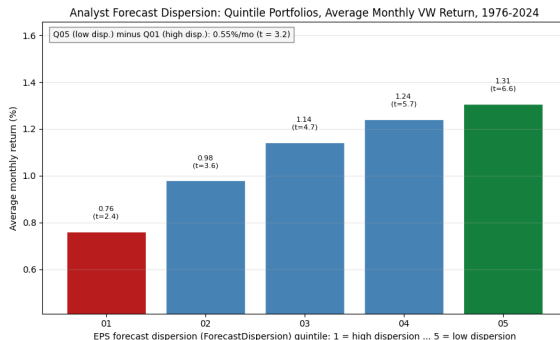
	(1)	(2)	(3)	(4)	(5)	(6)
<i>TURNOVER</i>	0.203			0.158		0.105
(<i>t</i> -stat)	(8.91)			(5.46)		(3.80)
<i>VOL</i>		21.76		14.12		13.99
(<i>t</i> -stat)		(5.38)		(2.97)		(3.05)
<i>FLOAT</i>			−0.302	−0.324		
(<i>t</i> -stat)			(12.62)	(12.31)		
<i>FIRMISSUE</i>					−1.302	−1.260
(<i>t</i> -stat)					(25.03)	(23.53)
<i>BROKERAGEISSUE</i>					0.496	0.419
(<i>t</i> -stat)					(15.84)	(12.60)

Notes: This table includes the regression results of daily warrant closing prices on *TURNOVER* (daily warrant turnover) in column 1, on *VOL* (5min warrant return volatility) in column 2, on *FLOAT* (the total number of warrants outstanding, in billions) in column 3, on *TURNOVER*, *VOL*, and *FLOAT* together in column 4, on *FIRMISSUE* (the total amount of warrants issued by the underlying firm from [Table 1](#) in billions) and *BROKERAGEISSUE* (net total warrant issued by brokerage firms, in billions) in column 5, and on *TURNOVER*, *VOL*, *FIRMISSUE*, and *BROKERAGEISSUE* together in column 6. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike) and include warrant maturity fixed effect. The *t*-statistics, in the parentheses, are adjusted for heteroskedasticity and correlation within a trading day.

Xiong and Yu (AER 2011)

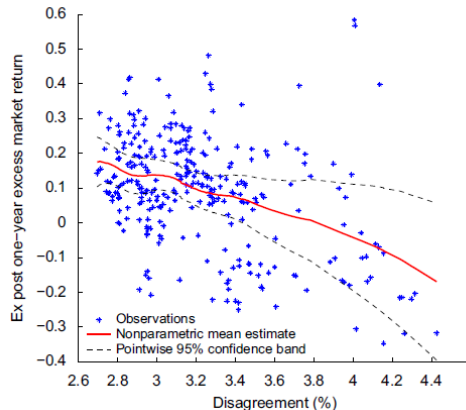
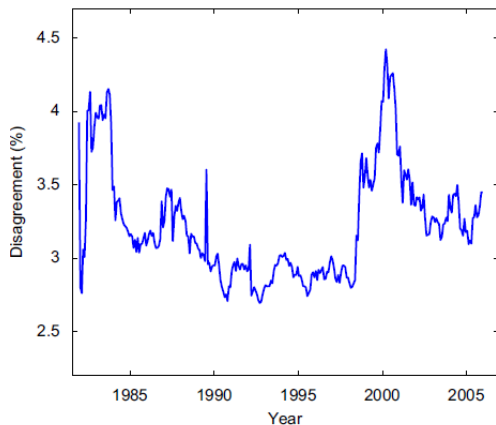
Heterogeneous Beliefs and Cross-Sectional Stock Returns

- Cross-sectional prediction of models of heterogeneous beliefs with short-sale constraints:
- Assets that people disagree about more should be more overpriced
- Diether, Malloy, and Scherbina (JF 2003) test the model's prediction using **analyst forecast dispersion** as a proxy for disagreement
- They find that stocks with higher analyst forecast dispersion earn lower future returns



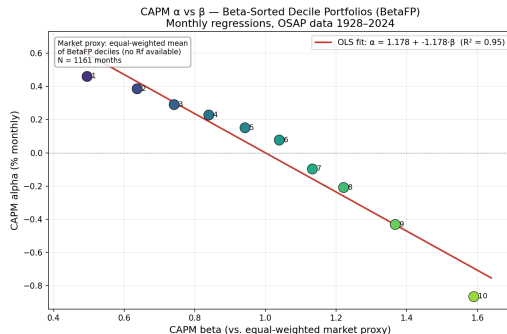
Heterogeneous Beliefs and Aggregate Stock Market

- Yu (2011) tests this prediction for the aggregate stock market
- Aggregate disagreement is measured as the value-weighted average of stock-level dispersion



Heterogeneous Beliefs and Risk-Return Tradeoff

- Another important application: risk-return tradeoff
- Standard theory: higher risk $\Rightarrow$ higher expected return
- But empirically, the relation between risk and return is often negative
- Heterogeneous beliefs models can explain this anomaly



Heterogeneous Beliefs and Risk-Return Tradeoff

- Heterogeneous beliefs can also explain the *high beta – low alpha* puzzle
- Hong and Sraer (2016): Disagreement about aggregate cashflows + short-sale constraints

$$d_i = \beta_i(\mathbf{Z}) + \mathbf{e}_i$$

- ① high β assets are more sensitive to disagreement about aggregate cashflows than low β assets
- ② Macro-disagreement: optimists $\mathbf{Z} + \lambda$, pessimists $\mathbf{Z} - \lambda$
- ③ high β assets experience more divergence of opinion: optimist $\beta_i(\mathbf{Z} + \lambda)$ and pessimists $\beta_i(\mathbf{Z} - \lambda)$
- ④ then overpricing of high β assets due to short-sale constraints

⇒ CAPM holds when aggregate disagreement is low

⇒ But high disagreement leads to inverted-U shape: initially increasing with beta and then decreasing

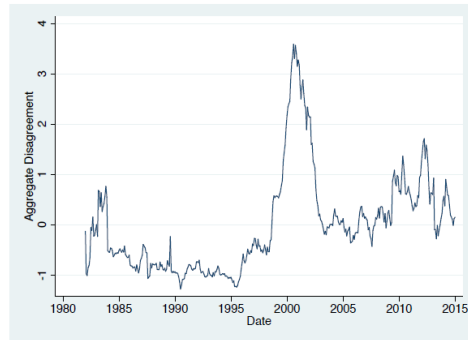
- β *amplifies aggregate disagreement*

Heterogeneous Beliefs and Risk-Return Tradeoff

- Empirical Predictions of Hong and Sraer (2016):
- Upward sloping SML when aggregate disagreement is low
- Inverted U-Shape of SML when disagreement is high
- Stock-level disagreement increases with beta, especially when aggregate disagreement is high

Figure 1: Time-series of Aggregate Disagreement

Note: Sample Period: 12/1981-12/2014. Sample: CRSP stock file excluding penny stocks (price < \$5) and microcaps (stocks in bottom 2 deciles of the monthly size distribution using NYSE breakpoints). Each month, we calculate for each stock the standard deviation of analyst forecasts on the stock's long run growth of EPS, which is our measure of stock-level disagreement. We also estimate for each stock i $\hat{\beta}_{i,t-1}$, the stock market beta of stock i at the end of the previous month. These betas are estimated with a market model using daily returns over the past calendar year and 5 lags of the market returns. Aggregate Disagreement is the monthly $\hat{\beta}_{i,t-1}$ -weighted average of stock-level disagreement.



Heterogeneous Beliefs and Risk-Return Tradeoff

Figure: Average 3-months excess returns for equal-weighted beta decile portfolios in low and high aggregate disagreement months.

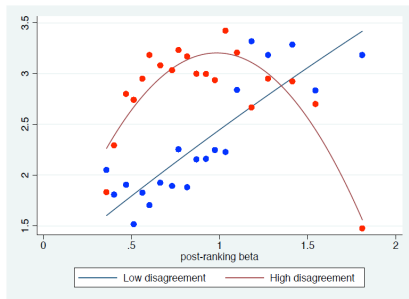
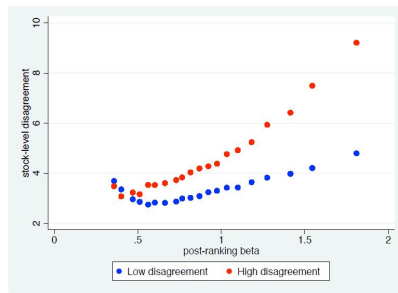
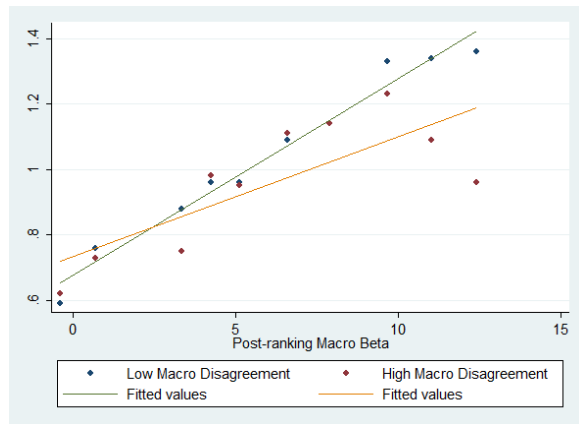


Figure: Equal-weighted average of dispersion of analyst earnings forecasts by beta deciles during low and high aggregate disagreement months.



Heterogeneous Beliefs and Risk-Return Tradeoff

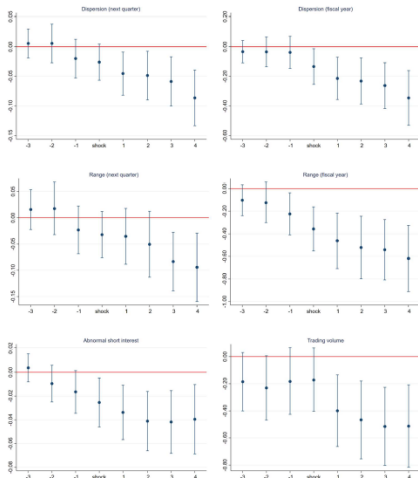
- The same intuition can explain the pricing of macroeconomic factors: my sole-authored work (Li, RAPS 2016)
- Using forecast dispersion from the Survey of Professional Forecasters, high disagreement about a macro factor predicts *lower* returns for stocks with high loadings on that factor



The Causal Effect of Disagreement on Asset Prices

- Identifying exogenous shocks to investor disagreement is still challenging
- Chang, Hsiao, and Ljungqvist (JF 2022) exploit the staggered implementation of EDGAR by SEC, which induces a reduction in information acquisition cost
 - Before EDGAR, investors could access firms' mandatory filings only at high cost, either by subscribing to commercial data providers or by physically visiting the SEC's reference rooms
 - Beginning in April 1993, the SEC required U.S. firms to file their mandatory disclosures electronically through the EDGAR system
 - SEC randomly assigned firms to one of ten implementation waves, thereby staggering inclusion in EDGAR over a three-year period between 1993 and 1996
- EDGAR inclusion helps resolve disagreement around information events, leading to price correction and lower price crash risk

The Causal Effect of Disagreement on Asset Prices



	Skewness (<i>NCSKEW</i>) (1)	Down-to-up volatility (<i>DUVOL</i>) (2)	Extreme negative returns, 0.01% (<i>CRASH001</i>) (3)
Quarter of EDGAR inclusion	0.007 0.022	0.000 0.014	-0.003 0.008
Next four quarters	-0.049** 0.020	-0.031** 0.012	-0.027*** 0.007
Controls?	yes	yes	yes
Calendar quarter FE?	yes	yes	yes
Fiscal quarter FE?	yes	yes	yes
Firm FE?	yes	yes	yes
R-squared	17.4%	19.1%	15.8%
Pre-trends (p-value)	0.562	0.141	0.957
No. of firms	3,366	3,366	3,366
No. of firm-quarters	28,652	28,652	28,652

Biased Beliefs: Concerns

- Do people simply learn their way out of biased beliefs?
- Only to a limited extent
- Many of the most persistent behavioral biases are useful mental shortcuts for human evolution
 - “Loss aversion” increases survival probability
 - Overconfidence increases effort and trust within your community
- Do highly trained professionals exhibit these biases?
 - Yes, professionals sometime do worse
 - CEOs and founders are more overconfident than the average person
- Do people exhibit these biases even when stakes are high?

Stakes and Investor Biases

- Sui and Wang (2025 JFE): investors do better when they care less
- Unique setting where investors trade stocks in real accounts using their own money and simultaneously in a simulated setting
- Within-investor estimation shows that investors exhibit stronger biases and perform worse in higher-stakes real accounts than in lower-stakes simulated accounts
- Explanation: individuals carry incorrect mental models and greater attention/effort leads to the more careful application of those incorrect models

Table 6

Differences between the real and simulated accounts. In this table, we report differences between real and simulated accounts. See [Section 4.1](#) for the bias definitions. For each type of bias we report the mean of the real accounts, the mean of the simulated accounts, and the mean of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column “N” we report the number of investors.

	N	Real	Simulated	Difference		
		Mean	Mean	Mean	<i>t</i> -stat	Wilcoxon <i>p</i>
Disposition effect	4371	4.163	0.409	3.674	(32.43)	0.000
Lottery preference	4369	6.491	4.264	2.228	(13.00)	0.000
Extrapolation	4373	15.599	12.66	2.939	(6.49)	0.006
Underdiversification	4394	-1.006	-1.473	0.306	(32.67)	0.000
Local Bias	1739	0.630	0.645	-0.015	(-0.64)	0.510
Turnover	4306	5.960	0.839	5.121	(47.44)	0.000
Performance	4375	-0.656	-0.148	-0.509	(-4.95)	0.001